

Unit 14: Applications of Data Warehousing and Data Mining

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Objectives

After this unit, you will be able to

- Understand the importance of data warehouses in financial data analysis and the retail industry.
- Know the importance of data warehouses in the Indian Railway reservation system and other industries.
- Learn the importance of weather forecasting in real life.

Introduction

Online transaction processing (OLTP) systems address an organization's operational data needs, which are critical to the day-to-day operation of a business. However, they are not well suited to sustaining decision support or business questions that managers are frequently faced with. Analytics, such as aggregation, drill-down, and slicing/dicing of data, are best supported by online analytical processing (OLAP) systems for such questions. By storing and managing data in a multidimensional manner, data warehouses aid OLAP applications. Extract, Transfer, and Load (ETL) tools are used to extract and load data into an OLAP warehouse from numerous OLTP data sources. The requirement for a data warehouse is driven by a business need and a business strategy. A data warehouse is a foundation for sophisticated data analysis; it aids business decision-making by allowing managers and other company users to analyze data and conduct analysis more efficiently.

14.1 Case Study of data warehouse in Financial Data Analysis

Financial Data Analysis

A financial data warehouse built with cutting-edge technology can help us improve the quality of our data and obtain insights into client behavior. As a result of these new insights, clients frequently increase the performance of their marketing campaigns and loyalty programs. Our financial analytics services are business-driven, and our professionals take the time to understand our specific needs and goals. Our consulting services assist financial companies in achieving the following goals:

- Make more informed decisions that are simple to access.
- Introduce new products to the marketplace.

- There are an increasing amount of restrictions.

Financial data warehouses work in the same way as regular data warehouses do. After data is acquired and put in a data warehouse, it is structured into a specific schema that categorizes the data. When the data needs to be analyzed, this allows for easy access.

A financial data warehouse using the latest technologies can increase the quality of our data and help us to gain insights into customer behavior. Clients often improve the effectiveness of their marketing campaigns and loyalty programs as a result of these new insights. Our financial analytics services are business-driven and our experts spend time establishing our requirements and task.

The Challenge

Every client stakeholder wanted an answer to the same question: How are sales going? There was no single location where executives, finance managers, and sales managers could get a comprehensive picture of the sales team's operations and evaluate key data such as present performance, prior performance, and pipeline. The client's IT department was having trouble keeping its governance and planning initiatives on track. They had taken measures to centralize the numerous data sources they needed to deal with and get the ball rolling on accurate, actionable reporting on those assets, but their infrastructure was outdated and no longer capable of handling the increased information demands. They engaged Ironside to help them upgrade their old data warehouse to a more current system that would help them achieve three important department goals:

- Remove warehouse infrastructure from its existing state of obsolescence.
- For more accurate referencing of past events, save historical point-in-time data.
- Increase query efficiency and enable more timely analyses by addressing the pain points that exist between the data warehouse and the Cognos BI reporting layer.

The Journey

The Ironside Information Management team agreed to collaborate with the client's database engineers to make the move to a modern data warehouse a success. The Ironside resources assigned to the project outlined a full-scale migration and redesign plan that would bring about 20 tables from a variety of data sources, including TM1, A-Track, Remedy, Oracle GL, and Excel, into an IBM PureData for Analytics (Netezza) implementation capable of meeting and exceeding the client's requirements, based on discovery conversations with IT leadership. Ironside was also tasked with reengineering both the ETL processes required to transport and transform all of the numerous information streams, as well as the reporting layer that makes that information available for analysis, as part of the migration.

So far, Ironside has completed the following phases:

- For the new data warehouse, They gathered requirements and use cases.
- The existing warehouse logic was documented, including all data extraction, transformations, schedules, and so on.
- Served as the solution architect for a data warehouse redesign proof of concept.
- Set specifications for new ETL operations to feed the POC design with the client's database engineers.
- The reporting layer was rebuilt to work in the new environment.
- Using the POC system, they tested data handling and report output.

The Results

The initial findings coming out of the concepting and testing phases of the project are very promising, and Ironside's team is confident that the final solution will deliver the modern data handling functionality that the client needs to continue their success.

The positive outcomes are:

1. Improved Cognos reporting usability and efficiency.
2. Support time comparison reporting, such as comparing data from the previous month to the current day.

3. During testing, a performance boost of roughly 60 times faster than achievable in the old environment was noticed.

Ironside and the customer are now making modifications and ready to roll out the full-scale data warehouse solution, using this strong proof of concept as a springboard. The IT staff will be able to easily fulfill the severe expectations of the financial services industry and give the sort of answers that will propel the company forward with this level of data performance at their disposal.

Advantages of Data Warehouses in the Financial Industry

Following are some of the advantages:

1. Analytic Types

Many businesses are interested in using financial data for a range of analytics purposes. Predictive and real-time analytics are the most popular types of analytics in finance. The goal of predictive analytics is to find patterns in financial data to anticipate future events. Real-time analytics is employed in a variety of applications, including consumer intelligence, fraud detection, and more.

2. Capturing Customer Data

Customers today use various channels on multiple devices, making data collection more challenging and likely to become more complex in the future. Data warehouses enable businesses to record every connection with a consumer, providing them with unprecedented insight into what motivates them.

3. Personalization

Many financial institutions have pushed to engage more with clients to stay competitive. One approach to achieve this is to send more tailored messages, which is possible thanks to data science. The ability to create more tailored interactions means that you can reach out to customers at the most appropriate times and with the most effective message.

4. Risk Management

Many financial institutions have pushed to engage more with clients to stay competitive. One approach to achieve this is to send more tailored messages, which is possible thanks to data science. The ability to create more tailored interactions means that you can reach out to customers at the most appropriate times and with the most effective message.



Example: Data warehouses are largely utilized in the investing and insurance industries to assess consumer and market trends, as well as other data patterns. Data warehouses are critical in two key sub-sectors: forex and stock markets, where a single point discrepancy can result in enormous losses across the board. Data warehouses are typically shared in these industries and focus on real-time data streaming.

14.2 Data Warehouse in Retail Industry

The retail industry collects a large amount of database and information on sales and customer shopping history. The quantity and quality of data and information collected continue to rapidly, especially due to the increasing ease, availability, and popularity the business conducted on the web, or retail industry provides a rich source for data mining. Retail database mining can help identify :

- Customer behavior,
- Discover customer shopping trends,
- Improve the quality of customer service and patterns,
- Achieve better customer retention and satisfaction and feedback,
- Enhance goods consumption ratios design more effective goods transportation and distribution policies and reduce the cost of business.

Why the Retail Industry Needs Data Warehousing and Analytics?

Every business, including retail, is being disrupted by the digital revolution, which is offering tremendous new opportunities. AI, robotics, blockchain, and data analytics are all transforming the way the retail industry interacts with and serves its customers. Retailers are utilizing data warehousing and analytics in novel and unique ways, all with the goal of better serving their customers and improving their shopping experience. Data warehousing and analytics have various

advantages, including increased operational efficiency, improved customer experience, and customer loyalty and retention.

The following are four ways that data warehousing and analytics are transforming the retail industry:

1. A Data Warehouse Saves Time

Data warehousing allows you to develop a centralized system for operations, data collection, and analysis. This eliminates the requirement for manual data exporting to excel sheets for reporting across corporate functions. Business users may get vital information from a variety of sources in one place. They would not squander time obtaining information from different sources.

2. Enhanced Business Intelligence

Data warehousing and analytics gather information from several departments and show it in reports and dashboards. Furthermore, the dashboard's self-service capabilities allow users to slice and dice data, execute drill-down and drill-up tasks, and see data at both a holistic and granular level. This allows business users to gain key business insights and make quick, well-informed decisions to improve operational efficiency and growth.

3. Demand Forecasting and scaling of operations

Retailers require demand forecasting because their scope of operation does not remain constant throughout the year. It varies dramatically depending on the season. For retailers, for example, the holiday season accounts for a significant portion of annual revenues. As a result, it is vital that they scale up to meet higher demand during peak seasons and then scale down to eliminate excess inventory when sales return to normal.

4. Better understanding of customers

Retailers can employ data warehousing and analytics to acquire a better understanding of their customers' behavior, tastes, and buying habits, and use the information for personalized offers, contextual marketing, and event planning shop design and aesthetics.

Data warehousing and analytics have become important to any retailer's success as the retail business grows and becomes more competitive. To meet business goals and generate important customer insights, merchants must be able to connect the proper, relevant data.



Notes: The Data Warehouse Data Model for Retail and Distribution (Retail DWH model ®) is a standard industry data warehouse model that covers traditional BI requirements, regulatory needs, and Big Data Analytics requirements for retailers and wholesalers.

Background of Apex Supermarket

Apex Grocery is a supermarket chain with 50 locations across the United States, including the West Coast, East Coast, Midwest, and Southwest. Georgia, New York, and Virginia are on the East Coast. Illinois and Kansas are in the Midwest. Texas is in the southwest.

Traditional System Limitation

- There is currently a lack of ability to make strategic decisions.
- No drill-down or roll-up reports are possible.
- It is not possible to conduct multi-dimensional analysis and make decisions.

Why a data warehouse

Allow Decision Support

1. Provide an information system for executives to use while making strategic decisions, such as:

- What is the most popular time for school supplies to be purchased?
- Which racks must school supplies be put to attract clients during the "school season"?

2. Establish an information system for executives to use in making strategic decisions, such as:

- Where in the store do perishable commodities such as fruits and vegetables need to be moved quickly?

- How long can meat meals be kept on the rack at Apex's Illinois location?

3. Roll-up and drill-down capabilities in the Central Repository provide sales by category, region, store, and more.



Example: For distribution and marketing, data warehouses are commonly employed. It also aids in the tracking of items, client purchasing patterns, and promotions, as well as for deciding pricing strategy.

14.3 Data Warehouse in Railway Reservation System

A data warehouse is a centralized storage location for all or large portions of the data collected by an organization's numerous business systems. Bill Inmon invented the phrase. The term "information warehouse" is sometimes used by IBM.

The passenger reservation system (PRS) is an OLTP system with a large number of concurrent reservation users actively adding and changing data. As the online PRS generates an increasing volume of data, the need to analyze data is becoming more and more by higher management for

1. Managing and utilizing the existing resources.
2. Creating new resources for the convenience of passengers.

When railway management tries to examine the data, however, so many of these issues often prevent it from being done:

- Railway executives (and even railway programmers) are unable to develop ad hoc inquiries.
- The application database is fragmented over several servers around the country, making it difficult for railway users to locate data in the first place.
- Ad-hoc querying of PRS OLTP systems is prohibited by database administrators to prevent analytical users from running queries (e.g., how many passengers are booked for a specific journey Day) that slow down mission-critical PRS databases.

Aim of study

We have recorded all of the information on the trains scheduled and the users booking tickets, as well as the condition of the trains, seats, and so on, in our study of the railway reservation system. This database is useful for applications that allow customers to book train tickets and check train details and status from their home, avoiding the inconvenience of having to go to the railway station for every question. Passengers can use the Railway Reservation System to inquire about:

1. The trains that are available based on their origin and destination.
2. Ticket booking and cancellation, as well as inquiring about the status of a booked ticket, and so on.
3. The goal of this case study is to design and create a database that stores information about various trains, train status, and passengers.

The major goal of keeping a database for the Railway Reservation System is to limit the number of manual errors that occur during ticket booking and cancellation, and make it easy for customers and providers to keep track of information about their clients as well as available seats. Many flaws in manual record-keeping can be eliminated due to automation.

The data will be obtained and processed at a rapid pace. The suggested system could be web-enabled in the future to allow clients to make numerous inquiries about trains between stations. As a result, a variety of issues arise from time to time, and they are frequently involved in consumer disputes. Some assumptions have been made to implement this sample case study, which is as follows:

1. The number of trains is limited to five.
2. The reservation is only available for the following seven days from the present date.
3. There are only two types of tickets available for purchase: AC and General.
4. There are a total of 10 tickets available in each category (AC and General).

5. The maximum number of tickets that can be given a waiting status is two.
6. The time between halt stations and their reservations is not taken into account.

Process

A train list must be kept up to date. Passenger information must be kept up to date. The passenger's train number, train date, and category are read throughout the booking process. An appropriate record from the Train Status is fetched based on the values provided by the passenger. If the desired category is AC, then the total number of AC seats and the number of booked AC seats are compared to find whether a ticket can be booked or not.

It can be checked in the same way for the general category. Passenger details are read and stored in the Passenger database if a ticket may be booked. The passenger's ticket ID is read during the cancellation procedure, and the Passenger database is checked for a matching record. The record is erased if it exists. After deleting the record (if it is confirmed), the Passenger table is searched for the first record with a waiting status for the same train and category, and its status is updated to confirm.

CRIS: The Need

For the following reasons, it was thought that a separate organization would be better prepared to take over all computer activity on IR:

1. To prevent individual railways from duplicating their efforts.
2. To ensure that computer hardware and software on the railways are all the same.
3. To design and create big railway applications that necessitate higher degrees of expertise, faster decision-making, and system-wide applicability.
4. To protect the organization from the day-to-day operations of the railways so that its goals are not forgotten.
5. The necessity for a collaborative effort between railways and computer specialists who are best qualified to build computer applications for railways.
6. Expertise development in highly specialized disciplines such as operations research, simulation, expert systems, CAD/CAM, and process control is required.
7. A greater degree of flexibility is required to keep up with rapidly changing technology.

14.4 Applications of Data Warehouse

The following are the various applications of data warehouse in various industries:

Government and Education: Data warehouses are used by the government to preserve and analyze tax records, health policy records, and their related providers, and the state's data warehouse is also related to the criminal law database. Patterns and trends, which are the outcome of analyzing historical data linked with prior criminals, are used to anticipate criminal activity.

Healthcare: All of their financial, clinical, and employee data is loaded into warehouses, which allows them to strategize and anticipate outcomes, track and analyze customer feedback, make patient reports, and exchange data with tie-in insurance companies, medical aid agencies, and so on.

Hospitality Industry: Hotel and restaurant services, automobile rental services, and vacation home services account for a large percentage of this industry. They employ warehousing services to create and assess their ad and marketing programs, which target customers based on their feedback and travel patterns.

Insurance: Apart from keeping track of existing participants, the warehouses are largely used to evaluate data patterns and customer trends. Warehouses can also be used to create customized consumer offers and promotions.

Manufacturing and Distribution Industry: This industry is one of a state's most important sources of revenue. A manufacturing company must make several make-or-buy decisions that can have a significant impact on the industry's future, which is why they use high-end OLAP tools as part of data warehouses to forecast market changes, analyze current business trends, detect warning conditions, track marketing trends, and, ultimately, to predict market changes. Data warehouses are used to handle the supply chain management of products in distributions.

Telephone Industry: The telephone industry uses both offline and online data, resulting in a large amount of historical data that must be aggregated and integrated.

Apart from those processes, a data warehouse is required for the study of fixed assets, the study of customer calling habits for salespeople to push advertising campaigns, and the tracking of customer inquiries.

Transportation Industry: Client data is recorded in data warehouses in the transportation industry, allowing traders to experiment with target marketing, where marketing campaigns are created with the needs of the customer in mind. They are used in the industry's internal environment to monitor customer feedback, performance, crew management on board, and customer financial information for pricing strategies.

14.5 Data Warehouse in Weather Forecasting

Weather forecasting has traditionally relied on huge, complicated physics models that take into account a variety of atmospheric circumstances over a lengthy period. Because of weather system perturbations, these conditions are frequently unstable, causing models to make erroneous forecasts. We describe a weather prediction methodology in this study that uses historical data from several weather stations to train simple machine learning models that can produce meaningful forecasts for specific weather conditions soon in a short amount of time.

Weather conditions around the world change rapidly and continuously. In today's world, accurate forecasts are critical. We rely significantly on weather forecasts in all we do, from agriculture to industry, from travel to the daily commute. Because the entire world is affected by climate change and its consequences, it is critical to accurately predict the weather to maintain smooth and seamless movement as well as safe day-to-day activities.

Weather Forecasting

The first step in constructing a data warehouse is to gather forecaster knowledge to construct the subject system. Weather forecasting is the use of science and technology to anticipate the state of the atmosphere in a specific region. Temperature, rain, cloudiness, wind speed, and humidity are all factors to consider. Weather warnings are a type of short-range forecast that is used to protect people's lives.

According to forecasters, there are currently six categories of subjects in our subject system:

1. **Target Element:** The target element of the forecast, such as rainfall, temperature, humidity, and wind, must be included in the subject system.
2. **Index Data:** Some Index data are useful; they are usually flags or indicators of future weather, such as the surface wind direction in Wutaishan, the 500hPa height change in west Siberia, and so on.
3. **Statistical Data:** Subjects like the average of 5 stations, the maximum temperature gradient in someplace, minimum relative humidity in the last 5 days, and others should be included in the subject system to analyze statistical features in a region or period.
4. **Transformed Data:** orthogonal transformations, such as wavelet transformation. And data that has been filtered using lowpass, highpass, and other techniques.
5. **Weather System:** Even though the forecaster's knowledge is based on the study of model output, synoptic expertise is vital. As a result, the weather system should be part of the topic system. High, low, large gradient area, saddle area for scale element and convergence center, and divergence are all examples of weather systems.

The Forecasting Process

Making a weather forecast involves three steps:

- Observation and analysis
- Extrapolation to find the future state of the atmosphere.
- Prediction of particular variables.

Methods of weather Prediction

In Weather prediction, there are three methods available:

- **Synoptic Weather Prediction**
- **Numerical Weather Prediction**
- **Statistical Weather Prediction**

Synoptic Weather Prediction

It is the most basic and conventional method of weather forecasting. Until the late 1950s, this approach was still in use. The term "synoptic" refers to the observation of various weather elements at a certain period. To a meteorologist, a weather map that represents the atmospheric conditions at a specific time is a synoptic chart. Creating synoptic charts regularly necessitates a massive gathering and analysis of observational data from thousands of weather stations. Certain empirical laws have been formed based on years of careful study of weather charts. These criteria aid forecasters in forecasting the rate and direction of weather system migration.

Numerical Weather Prediction

The numerical method necessitates a significant amount of mathematics. Numerical Weather Prediction techniques are now used in modern weather forecasting (NWP). This strategy is based on the fact that atmospheric gases adhere to a set of physical laws. The theoretical models of the general circulation of the atmosphere are developed using a series of mathematical equations. These equations are used to describe how the atmosphere evolves. Certain meteorological factors, such as air movements, temperatures, humidity, evaporation at the ground, clouds, rain, snow, and interactions of air with the earth and oceans, are taken into account in these equations. The daily weather prediction model is one such thing. On mobile phones, we see these forecasts.

Statistical Weather Prediction

Along with numerical weather prediction computations, statistical methods are primarily utilized. These methods are frequently used in conjunction with numerical methods. Statistical methods rely on weather data from the past, assuming that the future will be similar to the past. The major goal of looking at historical weather data is to figure out which features of the weather are good predictors of future events. Correct data can be safely utilized to anticipate future conditions after these linkages have been established. This method can only predict overall weather. It's especially useful for forecasting just one aspect of the weather at a time.

How is your business linked to the weather?

- Knowing the accurate situation of the weather is an important element for individuals and organizations.
- Many businesses are directly or indirectly linked with weather conditions.
- For instance, agriculture relies on perfect weather forecasting for when to plant, irrigate and harvest.
- Similarly, other occupations like construction, airport control authorities, and many more businesses are dependent on the weather.
- With weather forecasting, your organization can work more accurately without any disturbance.

Given the wide and essential need for accurate forecasting of weather conditions, data intelligence is powered by AI techniques that leverage real-time weather feeds and historical data. Live weather feeds from different locations (Latitude/Longitude) are available. Temperature, water level, wind, and other sensors continuously transmitting data. Historical weather data are publically available. Weather maps (precipitations, clouds, pressure, temperature, wind, weather stations) provide necessary information which will further help in the forecasting process.

Advantages of weather forecasting for businesses

- People get warned earlier of what the weather will be like for that particular day.
- Help people to take appropriate precautions to stay safe in case of unwanted occurrences.
- With forecasting methods, companies can get better outcomes with the help of accurate predictions.
- Delivers visual forecasts by methods most companies prefer.

- Helps agricultural organizations in buying/selling livestock.
- Helps the farming industry in planting crops, pastures, water supplies.
- The best method for inventory management, selling strategies, and crop forecasts.
- It provides the business with valuable information that the business can use to make decisions about future business strategies.

Challenges

Weather forecasts still have their limitations despite the use of modern technology and improved techniques to predict the weather. Weather forecasting is complex and not always accurate, especially for days further in the future, because the weather can be chaotic and unpredictable. If weather patterns are relatively stable, the persistence method of forecasting provides a relatively useful technique to predict the weather for the next day. Weather observation techniques have improved and there have been technological advancements in predicting the weather in recent times. Despite this major scientific and technical progress, many challenges remain regarding long-term weather predictability. The accuracy of individual weather forecasts varies significantly.

Summary

- Every business, including retail, is being disrupted by the digital revolution, which is offering tremendous new opportunities.
- The major goal of keeping a database for the Railway Reservation System is to limit the number of manual errors that occur during ticket booking and cancellation. and make it easy for customers and providers to keep track of information about their clients as well as available seats.
- A data warehouse is a centralized storage location for all or large portions of the data collected by an organization's numerous business systems.
- Hospitality Industry employs warehousing services to create and assess their ad and marketing programs, which target customers based on their feedback and travel patterns.
- Weather forecasting is complex and not always accurate, especially for days further in the future, because the weather can be chaotic and unpredictable.
- Data warehousing and analytics have various advantages, including increased operational efficiency, improved customer experience, and customer loyalty and retention.

Keywords

Key attribute: The key attribute is the attribute in a dimension that identifies the columns in the dimension main table that is used in foreign key relationships to the fact table.

Integrated: A data warehouse is developed by combining data from multiple heterogeneous sources, such as flat files and relational databases, which consequently improves data analysis.

Data Mart: A data mart performs the same function as a data warehouse, but with a smaller scope. It could be tailored to a certain department or line of business.

Self Assessment Questions

1. Using _____ Clients often improve the effectiveness of their marketing campaigns and loyalty programs as a result of these new insights.

- A. Financial Data Analysis
- B. Data Analysis
- C. Cash data Analysis
- D. None

2. Financial data consulting services help financial businesses to:

- A. Make decisions more effectively and easy to access.

- B. Bring new products onto the market.
 - C. A growing number of regulations.
 - D. All of the Above
3. _____ collects large amount of database and information on sales and customer shopping history.
- A. Retail Industry
 - B. Finance Industry
 - C. Manufacturing Industry
 - D. Production Industry
4. Retail database mining can help identify
- A. Customer behavior
 - B. Discover customer shopping trends
 - C. Improve the quality of customer service and patterns
 - D. All of the Above
5. Which of the following is/are new-age technologies
- A. AI
 - B. Data Analytics
 - C. Blockchain
 - D. All of the Above
6. A _____ is a central repository for all or significant parts of the data that an enterprise's various business systems collect.
- A. Data warehouse
 - B. Data Mart
 - C. DataBase
 - D. None
7. _____ is an OLTP system characterized by large numbers of concurrent reservation user's actively adding and modifying data.
- A. The person reservation systems
 - B. The passenger reservation systems
 - C. The personal reservation systems
 - D. None
8. Railway reservation system stored all the information about
- A. Trains scheduled
 - B. Users booking tickets
 - C. Status of trains and seats
 - D. All of the Above
9. The Railway Reservation System facilitates the passengers to enquire about
- A. The trains available based on source and destination.
 - B. Booking and Cancellation of tickets
 - C. Enquire about the status of the booked ticket
 - D. All of the above

10. The main purpose of maintaining a database for the Railway Reservation System is
- A. To reduce the manual errors involved in the booking and canceling of tickets
 - B. Make it convenient for the customers and providers to maintain the data about their customers
 - C. Data about the seats available
 - D. All of the Above
11. _____ is the prediction of the state of the atmosphere for a given location using the application of science and technology.
- A. Weather forecasting
 - B. Temperature forecasting
 - C. Rain forecasting
 - D. Humidity forecasting
12. _____ are a special kind of short-range forecast carried out for the protection of human life.
- A. Weather Messages
 - B. Weather Warnings
 - C. Weather Information
 - D. None
13. On an everyday basis, many use _____ to determine what to wear on a given day.
- A. Rain forecasting
 - B. Temperature forecasting
 - C. Weather forecasting
 - D. None
14. Which of the following steps making a weather forecast
- A. Observation and analysis
 - B. Extrapolation to find the future state of the atmosphere.
 - C. Prediction of particular variables.
 - D. All of the Above
15. It is the traditional and basic approach adopted in weather prediction.
- A. Synoptic Weather Prediction
 - B. Numerical Weather Prediction
 - C. Statistical Weather Prediction
 - D. None

Review Questions

- Q1) What is the significance of weather forecasting in today's business? Explain the various methods used in weather prediction.
- Q2) Discuss the various applications of Data Warehouse by giving the example of each application.
- Q3) "Every business, including retail, is being disrupted by the digital revolution, which is offering tremendous new opportunities". Justify the statement by giving the appropriate example.
- Q4) Discuss a scenario that shows how a data warehouse is helpful in the railway reservation system.
- Q5) Elucidate the role of Data Warehouse in financial data analysis.
- Q6) Discuss the various challenges faced during weather forecasting.

Answers: Self Assessment

1	A	2	D
3	A	4	D
5	D	6	A
7	B	8	D
9	D	10	D
11	A	12	B
13	C	14	D
15	A		

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